

Hybrid 2D–3D Reconstruction of the 2011 Tohoku Tsunami: Earthquake Generation, Regional Propagation, Local Hydrodynamic Impact and Coastal-Defence Assessment

Abstract

Confidence in numerical assessments of coastal-defence performance under tsunami loading is strengthened by validation against independent observations of tsunami propagation and coastal impact.

Here, a one-way coupled 2D–3D framework is developed to reconstruct the 2011 Tohoku tsunami from coseismic generation to coastal impact.

Coseismic seafloor displacement defines the initial free-surface disturbance for a 2D nonlinear shallow-water model simulating propagation through a source-to-coast corridor.

Bathymetric and topographic datasets define the terrain, while absorbing layers and radiation conditions reduce spurious reflections at truncated open-ocean boundaries.

Extracted free-surface-elevation and depth-averaged velocity histories are mapped to time-dependent inlet conditions for a 3D unsteady Reynolds-averaged Navier–Stokes model with volume-of-fluid interface capture, simulating inundation and interaction with rigid defences and buildings.

Model performance is evaluated against observed arrival times, offshore waveforms, run-up heights and inundation extents using errors in timing, amplitude and spatial extent.

Expected outcomes are close quantitative agreement with event observations, physically consistent regional-to-local forcing, and resolved flow fields and hydrodynamic loads.

The resulting event-validated framework will enable comparative assessment of defence configurations under the historical event and systematically varied source scenarios.

Subsequent development will incorporate fluid–structure interaction and damage modelling to assess defence deformation, damage progression and failure margins under extreme tsunami loading.

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1 Introduction

The project develops a one-way coupled tsunami simulation workflow for the 2011 Tohoku event, linking coseismic source generation, regional propagation and local three-dimensional hydrodynamic impact. The completed G6 gate establishes the theoretical 2D–3D hybrid model and its real-event integration baseline. It does not claim observational validation or physical calibration.

The motivation is methodological: regional nonlinear shallow-water models are efficient for source-to-coast propagation, while local URANS–VOF simulations resolve free-surface interaction with coastal defences and buildings. Prior hybrid and comparative studies support this separation of roles [1, 2, 3]. The Tohoku event provides the source, observation and impact context [4, 5, 6].

This document consolidates the Research workstream into the authoritative project report. The Research workstream remains the evidence layer; this report records the accepted G6 formulation, traceability, verification status and the methodology required before calibration and validation can be executed.

2 Project Aims and Scope

The project aim is to reconstruct tsunami propagation and local coastal-defence loading in a traceable, event-grounded workflow. The current status is:

- theoretical 2D–3D model: complete;
- G6 implementation verification: complete;
- real-event integration: complete;
- observational validation: not yet complete;
- calibration methodology: defined in this report;
- physical calibration: not started;

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- FSI, damage/failure, porous official breakwater representation and ML optimisation: future work.

The report therefore uses status terms carefully. *Implemented* means present in repository or adopted backend configuration. *Numerically verified* means tested by unit, benchmark or accepted run evidence. *Real-event integrated* means the 2011 Tohoku source-to-Kamaishi workflow has been executed through the G6 replay. *Validated* and *calibrated* are reserved for future comparison against approved observations and subsequent parameter updates.

3 Tohoku Case Study and Data

The event source is the 2011 off-Pacific-coast Tohoku earthquake. Repository preprocessing uses approved finite-fault input, with USGS and published source reconstructions retained as authority records [4, 7, 8, 9]. Source sensitivity remains an early calibration target because Tohoku-specific studies show sensitivity to source geometry, kinematics and dispersive/model-form assumptions [10, 11, 12].

Terrain and bathymetry are treated as measured or fidelity/model-form inputs rather than free calibration variables. The G6 baseline uses ETOPO 2022 as the accepted available terrain source, while higher-resolution J-EGG500, GEBCO and GSI products remain documented candidates for later verification and validation work [13–16].

Observation sources are not yet approved for calibration execution. The project expects offshore waveforms, arrival times, run-up heights and inundation extents to be ingested under an observation-approval process. Mori et al. provide a principal post-event survey authority, but the survey type, spatial density and tide-correction provenance must be respected [5].

4 Hybrid System Architecture

The architecture is a one-way pipeline:

1. finite-fault source and terrain preprocessing;
2. Regional2D finite-volume propagation;
3. extraction of a nearshore coupling interface;
4. mapping to Local3D OpenFOAM replay boundary data;

5. local no-defence and rigid-barrier hydrodynamic runs;
6. evidence and provenance capture.

Table 1: Traceability classification for major mathematical components.

Component	Implementation route	Evidence and limitation
Regional NLSWE	Repository-native	Research authority in RES-MOD/RES-NUM; G6 prefix equivalence accepted.
Earthquake preprocessing	Repository-owned preprocessing	Approved finite-fault input; source alternatives remain calibration sensitivity candidates.
Corridor/geospatial preprocessing	Repository-owned	EPSG:32654 corridor and conditioned terrain; higher-resolution products remain post-G6.
2D to 3D coupling	Repository-native replay/mapping	Fixed accepted source window 245–545 s mapped to 0–300 s.
Local3D URANS–VOF	Adopted OpenFOAM Foundation 11 backend	Repository defines dictionaries and evidence; OpenFOAM internals are adopted, not repository-native.
Wall functions	OpenFOAM-adopted with project policy	Continuous Spalding baseline; formal wall/mesh convergence remains post-G6.
Boundary treatment	Project-defined OpenFOAM configuration	<code>open_ocean_damped</code> policy accepted by reflection benchmark and real replay.

5 Earthquake Tsunami Generation

The source operator maps finite-fault earthquake parameters into vertical seabed displacement. The final G6 baseline uses repository-owned preprocessing with

an approved finite-fault artifact; alternative source inversions are sensitivity candidates, not hidden calibration offsets.

$$\Delta z_b(x, y) = \mathcal{O}(\mathcal{E}) \quad (5.1)$$

The initial free-surface condition applies passive transfer of the seabed displacement to the water surface:

$$\eta(x, y, 0^+) = \eta_0 + \Delta z_b(x, y) \quad (5.2)$$

This is the accepted G6 source-transfer assumption. Dispersive or fully three-dimensional source-generation effects remain model-form sensitivity topics motivated by the Tohoku literature [10, 11].

6 Regional2D Governing Model

Regional propagation is represented by the nonlinear shallow-water equations in conservative variables:

$$\mathbf{U} = \begin{bmatrix} h \\ q_x \\ q_y \end{bmatrix} \quad (6.1)$$

The balance law is

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} + \frac{\partial \mathbf{G}}{\partial y} = \mathbf{S}. \quad (6.2)$$

The fluxes are

$$\mathbf{F}(\mathbf{U}) = \begin{bmatrix} q_x \\ q_x^2/h + \frac{1}{2}gh^2 \\ q_x q_y/h \end{bmatrix} \quad (6.3)$$

and

$$\mathbf{G}(\mathbf{U}) = \begin{bmatrix} q_y \\ q_x q_y/h \\ q_y^2/h + \frac{1}{2}gh^2 \end{bmatrix}. \quad (6.4)$$

The bed-slope source is

$$\mathbf{S}_b = \begin{bmatrix} 0 \\ -gh \partial z_b / \partial x \\ -gh \partial z_b / \partial y \end{bmatrix}. \quad (6.5)$$

Manning friction is applied as

$$\mathbf{S}_f = -gn^2 \frac{|\mathbf{q}|\mathbf{q}}{h^{7/3}}, \quad \mathbf{q} = \begin{bmatrix} q_x \\ q_y \end{bmatrix}. \quad (6.6)$$

The Coriolis parameter is

$$f = 2\Omega \sin \phi. \quad (6.7)$$

Coriolis is treated as a measured physical/model-form input. It is not a blind calibration knob.

7 Regional2D Numerical Formulation

The accepted repository implementation is finite-volume based. The numerical formulation follows the Research workstream’s conservative-update intent while the final implementation is defined by the repository code and G6 evidence rather than by any stale intermediate note.

For cell K_i , the semi-discrete update can be written as

$$\frac{d\mathbf{U}_i}{dt} = -\frac{1}{|K_i|} \sum_{e \in \partial K_i} |e| \hat{\mathbf{H}}_{ie} + \mathbf{S}_i. \quad (7.1)$$

Well-balanced, positivity-aware shallow-water finite-volume methods motivate the treatment of wetting, drying and bed slopes [17, 18]. The G6 status is implementation verification, not formal convergence closure; mesh and timestep convergence remain post-G6 verification work.

8 Kamaishi Corridor and Terrain

The G6 case constructs a Tohoku–Kamaishi corridor in projected coordinates. The accepted CRS is EPSG:32654. Terrain conditioning uses the available ETOPO 2022 source for the G6 baseline; this supports the theoretical hybrid closure but does not complete high-resolution local validation.

The nearshore coupling interface is tagged as a Regional2D export boundary. The accepted source replay window is 245–545 s in the Regional2D history. That

interval is shifted to the Local3D replay interval 0–300 s, with the major shifted peak at 240 s.

Bathymetry and topography are fidelity inputs. They may trigger model-form investigation when timing is correct but nearshore amplitude or inundation extent is wrong, yet they must not be adjusted without approved data provenance [13, 14].

9 Regional-to-Local Coupling

The coupling maps Regional2D free-surface elevation and depth-averaged momentum to the Local3D inlet. For a local inward normal $\mathbf{n} = (n_x, n_y)$ and tangent $\boldsymbol{\tau} = (\tau_x, \tau_y)$,

$$q_n = q_x n_x + q_y n_y \quad (9.1)$$

and

$$q_\tau = q_x \tau_x + q_y \tau_y. \quad (9.2)$$

The reconstructed velocities are

$$u_n = \frac{q_n}{h} \quad (9.3)$$

and

$$u_\tau = \frac{q_\tau}{h}. \quad (9.4)$$

Discrete discharge preservation is checked by comparing the Regional2D interface flux with the Local3D boundary flux:

$$Q_{2D} = \sum_{j=1}^{N_s} q_{n,j} \Delta s_j \approx \int_{\Gamma_{\text{in}}} \mathbf{u} \cdot \mathbf{n} \, dA = Q_{3D}. \quad (9.5)$$

The G5/G6 prefix-equivalence evidence demonstrates that the accepted G6 forcing prefix is identical to the accepted G5 prefix over 0–600 s.

10 Local3D URANS–VOF Model

Local3D hydrodynamics are implemented through the adopted OpenFOAM Foundation 11 incompressible two-phase URANS–VOF backend. The repository owns

the case-generation policy and validation wrapper; it does not reimplement Open-FOAM solver internals.

The local continuity equation is

$$\nabla \cdot \mathbf{u} = 0. \quad (10.1)$$

The two-phase URANS momentum equation is represented as

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) = -\nabla p + \nabla \cdot [(\mu + \mu_t) (\nabla \mathbf{u} + \nabla \mathbf{u}^T)] + \rho \mathbf{g}. \quad (10.2)$$

The VOF phase-fraction equation is

$$\frac{\partial \alpha}{\partial t} + \nabla \cdot (\alpha \mathbf{u}) + \nabla \cdot [\alpha(1 - \alpha) \mathbf{u}_c] = 0. \quad (10.3)$$

Mixture properties are

$$\rho(\alpha) = \alpha \rho_w + (1 - \alpha) \rho_a \quad (10.4)$$

and

$$\mu(\alpha) = \alpha \mu_w + (1 - \alpha) \mu_a. \quad (10.5)$$

The SST turbulence model is represented by transport equations for k and ω , with the adopted backend supplying the detailed coefficients [19]:

$$\frac{\partial k}{\partial t} + \mathbf{u} \cdot \nabla k = P_k - \beta^* k \omega + \nabla \cdot [(\nu + \sigma_k \nu_t) \nabla k] \quad (10.6)$$

and

$$\frac{\partial \omega}{\partial t} + \mathbf{u} \cdot \nabla \omega = \gamma \frac{\omega}{k} P_k - \beta \omega^2 + \nabla \cdot [(\nu + \sigma_\omega \nu_t) \nabla \omega] + D_\omega. \quad (10.7)$$

The eddy viscosity closure is

$$\nu_t = \frac{a_1 k}{\max(a_1 \omega, SF_2)}. \quad (10.8)$$

Barrier traction combines pressure and viscous stress:

$$\mathbf{t} = \boldsymbol{\sigma} \mathbf{n}_b = [-p \mathbf{I} + \boldsymbol{\tau}_v] \mathbf{n}_b. \quad (10.9)$$

The force is

$$\mathbf{F} = \int_{\Gamma_b} \mathbf{t} dA \quad (10.10)$$

and the moment is

$$\mathbf{M} = \int_{\Gamma_b} (\mathbf{x} - \mathbf{x}_0) \times \mathbf{t} dA. \quad (10.11)$$

The future impulse metric is

$$\mathbf{J} = \int_{t_1}^{t_2} \mathbf{F}(t) dt. \quad (10.12)$$

Local3D turbulence-model sensitivity is methodology-relevant [20, 21], but physical calibration has not started.

11 OpenFOAM Implementation

The accepted backend is OpenFOAM Foundation 11, executed through the image `docker.io/openfoam/openfoam11-paraview510:11`. The repository generates OpenFOAM dictionaries, boundary data, wall-function policies, timestep policies and evidence summaries. The solver authority remains OpenFOAM; the repository authority is the case-generation and acceptance contract.

The Local3D replay schema is 1.1.0. Legacy schema 1.0.0 remains retained for historical synthetic fixtures, while production cases use the `open_ocean_damped` boundary policy. The G6 accepted implementation includes no-defence and simple-rigid-barrier variants.

12 Boundary, Wall and Timestep Treatment

The production boundary policy includes a Regional2D-forced inlet, open outlet, lateral open-ocean boundaries, atmospheric top, terrain wall, barrier wall and isotropic damping zones. The reflection benchmark distinguishes reflective control behavior from production outlet and lateral behavior:

- reflective control: $K_r = 0.8199788627$, $RE = 0.6724000004$;
- production outlet: $K_r = 0.0799979378$, $RE = 0.006400000004$;
- production lateral: $K_r = 0.1000003891$, $RE = 0.009999999974$.

The wall-function baseline uses continuous Spalding treatment for turbulent viscosity, while `kqRWallFunction` and `omegaWallFunction` are retained for turbulence quantities. Runtime y^+ evidence is accepted for the G6 baseline. Formal wall-function/mesh convergence remains post-G6.

The timestep policy is accepted for the G6 baseline: repository configuration defines pre-run caps and minimum timestep acceptance, while OpenFOAM controls internal Courant and interface-Courant adaptation. Formal timestep convergence and an optional external rollback/retry supervisor remain post-G6, preserving issue #62.

13 G6 Verification and Model Closure

The frozen G6 baseline is:

- commit: 319d347644f0fe057ca2faa66ca5e324e10b5a21;
- OpenFOAM: Foundation 11;
- image: `docker.io/openfoam/openfoam11-paraview510:11`;
- replay schema: 1.1.0;
- production boundary mode: `open_ocean_damped`;
- accepted Regional2D source window: 245–545 s;
- Local3D replay: 0–300 s.

The metric-scope correction is part of the accepted evidence. The historical full-history normal-momentum maximum is $4.779893640553428 \text{ m}^2/\text{s}$ over 0–1800 s. The selected coupling-window normal-momentum maximum is $4.2421347278949835 \text{ m}^2/\text{s}$ over 245–545 s. These are not equivalent quantities because they are computed over different temporal scopes.

Both accepted Local3D cases reached 300 s. The no-defence case and the simple-rigid-barrier case passed runtime boundedness, final-time, boundary-data, VTK and probe/force evidence checks.

14 Validation Methodology

Validation will compare simulated signals η_s against approved observations η_o , without arbitrary timing shifts in the primary metrics. The point error is

$$e_i = \eta_s(t_i) - \eta_o(t_i). \quad (14.1)$$

The RMSE is

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_i e_i^2}. \quad (14.2)$$

The MAE is

$$\text{MAE} = \frac{1}{N} \sum_i |e_i| \quad (14.3)$$

and the bias is

$$\text{Bias} = \frac{1}{N} \sum_i e_i. \quad (14.4)$$

Peak amplitude and timing errors are

$$E_{\eta, \max} = \eta_{s, \max} - \eta_{o, \max} \quad (14.5)$$

and

$$E_{t, \max} = t_{s, \max} - t_{o, \max}. \quad (14.6)$$

A relative peak error is

$$E_{\eta, \text{rel}} = \frac{\eta_{s, \max} - \eta_{o, \max}}{\eta_{o, \max}} \quad (14.7)$$

and correlation is

$$r = \frac{\sum_i (\eta_s(t_i) - \bar{\eta}_s)(\eta_o(t_i) - \bar{\eta}_o)}{\sqrt{\sum_i (\eta_s(t_i) - \bar{\eta}_s)^2} \sqrt{\sum_i (\eta_o(t_i) - \bar{\eta}_o)^2}}. \quad (14.8)$$

Time-dependent diagnostics are

$$\text{RMSE}(T) = \sqrt{\frac{\sum_{t_i \leq T} e_i^2}{N(T)}}, \quad (14.9)$$

$$\text{MAE}(T) = \frac{\sum_{t_i \leq T} |e_i|}{N(T)}, \quad (14.10)$$

and

$$\text{Bias}(T) = \frac{\sum_{t_i \leq T} e_i}{N(T)}. \quad (14.11)$$

Cross-correlation lag may be recorded as a diagnostic, but physical timing remains unshifted for primary validation.

15 Calibration and Sensitivity Methodology

Calibration is hierarchical and diagnostics-led rather than blind optimisation. The formal target can be written as

$$\boldsymbol{\theta}^* = \arg \min_{\boldsymbol{\theta} \in \Theta} J(\mathbf{y}_s(\boldsymbol{\theta}), \mathbf{y}_o), \quad (15.1)$$

but parameter changes are only admissible when an error pattern, physical hypothesis and expected directional effect are recorded.

Table 2: Calibration parameter taxonomy.

Parameter or model choice	Class	Status
Earthquake source model	D	source sensitivity candidate
Source amplitude multiplier	C	conditional; only with source-evidence support
Manning n	C	regional friction calibration candidate
Bathymetric dataset	F	fidelity/model-form choice
Mesh resolution	V/F	verification first, fidelity second
Astronomical tide	M	measured physical input
Coriolis	M	measured/model-form physical input
Sponge/damping	V	numerical boundary verification parameter
Local3D turbulence intensity I	C	local inlet/turbulence candidate
Turbulence length scale L_t	C	local turbulence candidate
Wall roughness k_s	future C	future roughness calibration
Coupling vertical velocity profile	D/F	model-form sensitivity
Porous resistance parameters	future work	not in current calibration vector

Table 3: Cause–change–effect–rationale register examples.

Trigger	Hypothesised cause	Candidate change	Conditions
Offshore timing wrong	source geometry bathymetry error	investigate source model or terrain fidelity	do not tune friction first.
Offshore timing good but amplitude wrong	earthquake source amplitude/kinematics	source-model sensitivity	require source citation and observation approval.
Offshore waveform good but nearshore amplitude high	friction or bathymetric transformation	Manning n study	only after convergence checks.
Regional2D forcing accepted but 3D free surface wrong	coupling, turbulence or local geometry	vertical profile/turbulence/local geometry sensitivity	preserve forcing provenance.
3D surface credible but loads wrong	turbulence, roughness or geometry	turbulence/roughness geometry load investigation	arbitrary load multiplier.

The sensitivity coefficient is

$$S_{ij} = \frac{\theta_j}{y_i} \frac{\partial y_i}{\partial \theta_j}. \quad (15.2)$$

A centred finite difference estimate is

$$\frac{\partial y_i}{\partial \theta_j} \approx \frac{y_i(\theta_j + \Delta\theta_j) - y_i(\theta_j - \Delta\theta_j)}{2\Delta\theta_j}. \quad (15.3)$$

Parameter identifiability requires that multiple observables respond in distinguishable directions. Tohoku-specific literature motivates source representation as an early sensitivity target; project-specific sensitivity testing is still required before calibration [10, 11, 12].

Every calibration iteration is recorded as

$$\mathcal{C}_k = \{\theta_{\text{old}}, \theta_{\text{new}}, H_k, E_k, R_k\}, \quad (15.4)$$

where H_k is the physical hypothesis, E_k is the expected effect and R_k is the realised effect. Each run record must include run ID, parent run, baseline commit, parameter, old and new value, unit, reason, error pattern, hypothesis, expected effect, metrics before and after, actual effect, retain/revert decision and citation basis.

The following are numerical controls, not physical calibration variables: CFL limit, alpha-CFL limit, minimum timestep, solver tolerance, positivity limiter, dry-depth numerical tolerance, arbitrary time shift and arbitrary output amplitude multiplier. Mesh and timestep are selected through verification/convergence.

The porous-breakwater calibration literature is retained only for Future Work. Parameters such as α_p , β_p , ϕ and D_{50} may be introduced after an actual porous/rubble-mound barrier formulation is adopted [22].

16 Complete Computational Algorithm

Algorithm 1: End-to-end hybrid tsunami reconstruction and future calibration workflow.

- 1: Establish source provenance governance and approved observation registry.
- 2: Ingest candidate offshore, nearshore, run-up and inundation observations after approval.
- 3: Define numerical convergence plan for Regional2D and Local3D mesh/timestep controls.
- 4: Acquire and verify terrain, bathymetry and finite-fault source artefacts.
- 5: Transform event and corridor geometry to EPSG:32654.
- 6: Generate earthquake displacement Δz_b and initialise Regional2D free surface.
- 7: Solve the Regional2D finite-volume model with wet/dry, friction and boundary treatments.
- 8: Extract virtual gauges and nearshore coupling histories.
- 9: Perform source-model sensitivity after convergence and observation approval.
- 10: Calibrate Regional2D physical parameters only through documented hypotheses.
- 11: Execute Regional holdout validation without arbitrary time shifts.
- 12: Select accepted coupling window and project normal/tangential momentum.
- 13: Reconstruct 3D inlet velocity and VOF boundary data.

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- 14: Generate Local3D geometry, mesh, OpenFOAM dictionaries and provenance records.
 - 15: Run URANS–VOF no-defence and defence cases.
 - 16: Extract free surface, probes, forces and moments.
 - 17: Perform Local3D turbulence, roughness and geometry sensitivity as authorised.
 - 18: Calibrate Local3D physical parameters only after Regional forcing is accepted.
 - 19: Execute hybrid holdout validation.
 - 20: Archive HDF5/VTKHDF results, publication figures and provenance.
 - 21: Record retain/revert decisions and update the evidence ledger.
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The G6 work completed the implementation and real-event integration path through Local3D acceptance. The remaining algorithmic steps define validation, calibration, archive and publication work.

17 Current Results and Evidence

Accepted current evidence is limited to already executed G6 closure material:

- Regional2D real-event execution and fixed accepted coupling window;
- exact G5/G6 0–600 s forcing-prefix equivalence with maximum difference 0;
- reflection benchmark for production open-ocean outlet/lateral treatment;
- 300 s Local3D no-defence run;
- 300 s Local3D simple-rigid-barrier run;
- runtime boundedness, finite fields, probe output and VTK final-time output;
- wall-function and y^+ evidence;
- force and moment extraction for the rigid barrier case.

The current event reconstruction is not described as observationally calibrated. No final validation accuracy is claimed.

18 Reproducibility and Traceability

The calibration baseline commit is 319d347644f0fe057ca2faa66ca5e324e10b5a21. Major PRs include the pre-WBS baseline capture, Regional2D closures, Kamaishi delivery integration and PR #274 for G6 closure.

The replay schema is 1.1.0, the boundary mode is `open_ocean_damped`, the coordinate reference is EPSG:32654, the event source is the 2011 Tohoku finite-fault source, and the terrain source for the G6 baseline is ETOPO 2022. The accepted replay interval is 245–545 s in source time and 0–300 s in Local3D shifted time.

Verification evidence is stored in the repository and external simulation evidence trees. This section is documentation evidence toward reproducibility; it is not itself the complete reproducibility bundle.

19 Limitations

The G6 baseline is a theoretical and implementation closure, not a final predictive validation. Limitations include the use of available global terrain for G6, simplified rigid-barrier geometry, absence of observational calibration, no formal mesh/timestep convergence, and no official CAD-based porous or rubble-mound representation.

OpenFOAM provides the Local3D solver authority, so the repository evidence focuses on case-generation correctness, accepted policy, runtime boundedness and traceable outputs. OpenFOAM internals are not claimed as repository-native implementation.

20 Future Work

Near-term project completion work includes observational validation, physical calibration, mesh convergence, timestep convergence, observation-source approval, higher-resolution bathymetry, graded/adaptive mesh selection and publication-quality single-panel figures.

Longer-term framework extensions include HDF5/VTKHDF outputs, CPU/MPI/GPU acceleration, official Kamaishi CAD geometry, porous rubble-mound representation, buildings, FSI, damage/failure modelling and ML optimisation. Porous calibration parameters remain future work until a porous barrier formulation is introduced.

21 Conclusions

The project now has a formal deliverable structure, a shared citation bank and an authoritative consolidated technical report. The G6 baseline closes the theoretical hybrid model: real-event Regional2D forcing is coupled into Local3D OpenFOAM replay, no-defence and rigid-barrier cases reach 300 s, and boundary, wall-function and timestep evidence are accepted for the baseline.

The next scientific step is not further G6 implementation; it is controlled validation and calibration from the frozen baseline commit, after observation approval and numerical-convergence planning.

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